ModelIt!

Teacher's Guide

Vaccine Design Optimization

Modeling Immune Response Engineering from Antigen Selection to Population-Level Protection

Quick Reference

Grade Level	Grade 09L3
NGSS Standard	HS-LS1-4, HS-LS4-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS1-4, HS-LS4-2

Use a model to illustrate the role of cellular division in producing and maintaining complex organisms; construct an explanation based on evidence that the process of evolution primarily results from four factors.

Three-Dimensional Learning

Science Practice	**Developing and Using Models / Constructing Explanations** — Students develop an immune response model that connects molecular antigen-antibody interactions to cellular immune activation to population-level epidemiological outcomes, constructing explanations for vaccine design trade-offs.
Disciplinary Core Idea	**LS1.A Structure and Function / LS4.C Adaptation** — Immune system cells have specialized structures that enable antigen recognition, antibody production, and immunological memory; pathogen adaptation through mutation creates an evolutionary arms race that constrains vaccine design.
Crosscutting Concept	**Cause and Effect / Structure and Function** — Students trace causal chains from antigen molecular structure through immune cell recognition to antibody specificity and population-level protection, analyzing how pathogen structural evolution disrupts vaccine-induced immunity.

CAST Testing Alignment

MULTIPLE CHOICE:

A new SARS-CoV-2 variant emerges with 15 mutations in the spike protein receptor-binding domain. Based on your model, which vaccine design strategy would maintain the most protection against this variant?

CONSTRUCTED RESPONSE:

Using your immune response model, explain why a vaccine that produces very high antibody levels against one influenza strain might fail to protect against a new strain that emerges 6 months later. Reference at least three model components including Pathogen Mutation Rate, Cross-Reactivity, and Antigen Selection in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

This lesson introduces students to computational vaccine design — one of the most important applications of immunological modeling. Biotech skill focus: Multi-objective optimization against an evolving target. The COVID-19 pandemic demonstrated both the power of rapid vaccine development and the challenges of designing durable protection against a mutating pathogen.

How Vaccines Work: The Immune Response Cascade

Vaccines exploit the adaptive immune system's ability to learn and remember. When a vaccine antigen is injected, innate immune cells (dendritic cells, macrophages) engulf and process it, presenting peptide fragments on their surface via MHC (major histocompatibility complex) molecules. Helper T cells recognize these presented peptides and activate, releasing cytokines that drive B cell activation. Activated B cells with receptors that match the antigen undergo clonal expansion (rapid division) and affinity maturation (mutation and selection for tighter antigen binding). Some activated B cells differentiate into plasma cells that secrete large quantities of antibodies; others become long-lived memory B cells. Simultaneously, cytotoxic T cells are activated to kill cells displaying the target peptides. The entire process takes 1-2 weeks for a primary response but only 1-3 days for a secondary (memory) response — which is why vaccines work: the memory response is fast enough to clear the pathogen before it causes disease.

Antigen Selection: The Foundation of Vaccine Design

The choice of antigen determines everything else about a vaccine. An ideal antigen is: immunodominant (strongly recognized by diverse immune systems), essential for pathogen function (the pathogen cannot easily mutate it without losing fitness), surface-exposed (accessible to antibodies), and conserved across variants (mutations in this region are rare). In practice, these criteria often conflict. The SARS-CoV-2 spike protein receptor-binding domain (RBD) is highly immunodominant and surface-exposed but also rapidly mutating because it is under intense immune selection pressure. The spike protein's fusion machinery (S2 subunit) is more conserved but less immunodominant. The internal nucleoprotein is highly conserved but not accessible to antibodies (though T cells can target it). Computational modeling allows vaccinologists to evaluate thousands of potential antigen designs against simulated pathogen evolution, identifying targets that balance immunogenicity with conservation.

Adjuvants and Immune Amplification

Purified protein antigens alone often produce weak immune responses — the adaptive immune system needs 'danger signals' from the innate immune system to mount a full response. Adjuvants provide these danger signals. Aluminum salts (alum) — the oldest and most common adjuvant — create a depot effect that prolongs antigen exposure and activate inflammasome pathways. AS01B (used in the Shingrix shingles vaccine) combines a toll-like receptor agonist with a saponin to produce exceptionally strong immune responses. MF59 (an oil-in-water emulsion) enhances antigen uptake by dendritic cells. mRNA vaccines use the lipid nanoparticle delivery system as a built-in adjuvant because the mRNA itself activates innate immune sensors. The choice of adjuvant determines the magnitude, quality, and durability of the immune response — but stronger adjuvants also produce more local and systemic reactogenicity (injection site pain, fever, fatigue), which affects public acceptance and population-level uptake.

The Evolutionary Arms Race: Pathogen Mutation and Immune Escape

Vaccination creates a selection pressure on the pathogen: viral variants that can evade vaccine-induced immunity have a fitness advantage and become dominant. This is antigenic drift — the gradual accumulation of mutations in immune-targeted surface proteins. Influenza A virus mutates its hemagglutinin (HA) surface protein at a rate of approximately 4-6 amino acid substitutions per year, sufficient to require annual vaccine reformulation. SARS-CoV-2 demonstrated even faster antigenic evolution, with new variants of concern emerging every few months during the pandemic. The fundamental challenge for vaccine design is that the pathogen is not a static target — it is an evolving adversary that actively adapts to escape immune

pressure. Computational models that co-simulate vaccine-induced immune responses AND pathogen evolution are essential for designing vaccines that maintain effectiveness over time.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify nine components spanning vaccine design inputs (Antigen Selection, Adjuvant Strength, Pathogen Mutation Rate), immune response dynamics (Immune Cell Activation, Antibody Production Rate, Memory Cell Formation), and protection outcomes (Cross-Reactivity, Duration of Immunity, Population Coverage).

E **Establish Relationships**

Students determine that Antigen Selection, Adjuvant Strength, and Pathogen Mutation Rate are external inputs — two under the vaccinologist's control and one representing the evolutionary adversary — while Immune Cell Activation, Antibody Production Rate, Memory Cell Formation, Cross-Reactivity, Duration of Immunity, and Population Coverage emerge as biological responses.

V **Visualize & Model**

Students build an immune response model connecting vaccine inputs through immune cell activation and antibody production to protection outcomes, visualizing the trade-offs between specificity and breadth, potency and safety, and short-term and long-term protection.

E **Evaluate Outcomes**

Students run strain-specific, broadly-protective, and high-mutation-challenge scenarios to discover how antigen selection strategy determines vulnerability to pathogen evolution.

R **Revise & Extend**

Students add T Cell Response Quality, Delivery Platform Efficiency, or Immune Imprinting Effect to explore additional dimensions of vaccine design optimization.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"COVID-19 vaccines went from genetic sequence to emergency authorization in 11 months. The previous record was 4 years. How did computational modeling make this possible — and can we design vaccines that never need updating?"*

Watch For

Display the timeline comparison: traditional vaccine development (10-15 years) versus COVID-19 (11 months). Ask: What changed? What role did modeling play?

Slide 2: Learning Objectives

3 min

Say: *"Today you're vaccine designers. You'll learn why vaccine design is one of the hardest optimization problems in biology — because your target is literally evolving to escape you."*

Watch For

Pre-teach antigen and adjuvant. Use the analogy: an antigen is a 'wanted poster' that teaches your immune system what to look for; an adjuvant is the megaphone that makes sure the immune system pays attention.

Slide 3: Big Question

3 min

Say: *"You need a new flu vaccine every year because the virus mutates its surface proteins. But you only need one measles vaccine for life. What's different about measles? Can we make flu behave more like measles?"*

Watch For

Think-pair-share: Why do some vaccines last a lifetime while others need annual updating? What properties of the virus and the vaccine determine durability? Collect hypotheses.

Slide 4: LEVER Framework

2 min

Say: *"We're modeling the full cascade: from what you put in the vaccine to how the immune system responds to whether protection lasts when the virus mutates. And the virus IS mutating — that's the adversary in this model."*

Watch For

Preview LEVER steps. Emphasize the unique challenge: this model has an adversarial component (Pathogen Mutation Rate) that actively works against the vaccine design.

Slide 5: Activity 1: Components

8 min

Say: *"Three of these nine components are inputs. Two you control (Antigen Selection, Adjuvant Strength). One you DON'T control (Pathogen Mutation Rate) — and it's working against you. Sort them."*

Watch For

Guide sorting of external versus internal components. Emphasize the adversarial nature of Pathogen Mutation Rate as an uncontrollable external input that actively undermines vaccine effectiveness.

Slide 6: Activity 2: Connections

8 min

Say: *"The specificity-breadth trade-off is the central tension. A sniper rifle (strain-specific) or a shotgun (broadly protective)? Each has advantages. Each has costs."*

Watch For

Students map the immune cascade from antigen presentation through B cell and T cell activation to antibody and memory cell outputs. Trace how Antigen Selection choice propagates through Cross-Reactivity to Duration of Immunity.

Slide 7: Activity 3: Simulation

10 min

Say: *"Design your vaccine. Now watch what happens when the virus starts mutating. Which design strategy holds up best over 2 years of viral evolution?"*

Watch For

Students design their preferred vaccine strategy, predict performance, then simulate all three scenarios. The high-mutation challenge is the key insight — demonstrating how quickly strain-specific immunity erodes.

Slide 8: Discoveries

5 min

Say: *"You just discovered why we still don't have a universal flu vaccine after 80 years of trying — and why the mRNA platform that made COVID vaccines possible might finally crack this problem."*

Watch For

Connect model findings to real vaccine development challenges. Discuss mRNA platform advantages for rapid variant-updated vaccine design.

Slide 9: STEM Challenge

12 min

Say: *"The WHO just gave your lab a billion dollars. Design the universal flu vaccine that ends annual flu shots forever. What do you target? How do you prove it works? Go."*

Watch For

Groups design universal influenza vaccine strategies: antigen selection rationale, adjuvant choice, delivery platform, clinical trial design, and projections for cross-reactivity breadth. Present and defend.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Antigen Selection, Adjuvant Strength, Pathogen Mutation Rate
- **INTERNAL:** Immune Cell Activation, Antibody Production Rate, Memory Cell Formation, Cross-Reactivity, Duration of Immunity, Population Coverage

Relationships

- **Antigen Selection to Immune Cell Activation:** POSITIVE (+) — More immunogenic antigens (those that are well-recognized by human immune system components) produce stronger dendritic cell activation and more effective antigen presentation to T cells, amplifying the downstream adaptive response.
- **Adjuvant Strength to Immune Cell Activation:** POSITIVE (+) — Stronger adjuvants activate more innate immune cells and induce higher levels of co-stimulatory molecules and cytokines, amplifying antigen presentation and lowering the activation threshold for adaptive immune cells.
- **Immune Cell Activation to Antibody Production Rate:** POSITIVE (+) — Greater immune cell activation produces more activated helper T cells, which provide stronger signals for B cell clonal expansion and differentiation into antibody-secreting plasma cells.
- **Memory Cell Formation to Duration of Immunity:** POSITIVE (+) — More robust memory cell differentiation creates larger and more diverse pools of long-lived memory B cells and T cells, enabling faster and stronger recall responses for longer periods after vaccination.
- **Pathogen Mutation Rate to Duration of Immunity:** NEGATIVE (-) — Faster pathogen mutation generates variants that escape vaccine-induced immune recognition more quickly, eroding the effective Duration of Immunity even when memory cells remain functional — because the virus changes, not the memory.
- **Cross-Reactivity to Population Coverage:** POSITIVE (+) — Broader cross-reactivity means the vaccine provides at least partial protection against more pathogen variants, increasing the fraction of the population that encounters a variant covered by their vaccine-induced immunity.

Key Discoveries

- There is a fundamental trade-off between vaccine specificity and breadth — vaccines targeting strain-specific epitopes generate stronger antibody responses but are vulnerable to antigenic drift, while vaccines targeting conserved regions are more durable but less potent
- Adjuvant strength has diminishing returns — moderate adjuvant produces significantly better immune responses than no adjuvant, but very strong adjuvant increases side effects faster than it increases protection, reducing real-world uptake and Population Coverage
- Memory Cell Formation is the most critical component for long-term protection — even when circulating antibodies wane, robust immunological memory enables rapid recall responses that prevent severe disease upon re-exposure
- Population Coverage is limited by human genetic diversity — HLA (human leukocyte antigen) variation means the same vaccine antigen is presented differently to immune cells in different people, making universal vaccine design for genetically diverse populations inherently challenging

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Vaccines give you a weakened version of the disease"	While some historical vaccines used weakened (attenuated) live virus, most modern vaccines do NOT contain any live pathogen. Subunit vaccines contain only purified proteins. mRNA vaccines contain instructions for the body to make one viral protein (not the whole virus). Inactivated vaccines contain killed virus that cannot replicate. None of these can cause the disease they protect against. The immune system responds to the molecular shape of the antigen, not to the disease process itself.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"If a vaccine doesn't prevent infection 100%, it doesn't work"	Vaccines provide a SPECTRUM of protection, not a binary shield. Even when a vaccine doesn't prevent infection entirely, it typically: reduces symptom severity by 70-90%, reduces hospitalization by 80-95%, reduces death by 90-99%, and reduces transmission by 40-60%. The model's Population Coverage component shows that even partially effective vaccines dramatically reduce disease burden at the population level through herd immunity effects and reduced transmission.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Natural immunity is always better than vaccine immunity"	Natural infection sometimes produces stronger immunity than vaccination — but at the cost of suffering the actual disease, with its risks of hospitalization, long-term complications, and death. The model shows that vaccines can be designed to optimize specific aspects of immunity (like Memory Cell Formation and Cross-Reactivity) that natural infection may not optimally stimulate. Adjuvants and delivery platforms can direct the immune response toward the most protective pathways. Furthermore, vaccines can target conserved epitopes that natural infection may not prominently expose to the immune system.